House Price Prediction

Kaggle House Prices: Advanced Regression Techniques — exploring when linear models fall short and whether combining linear and nonlinear models improves prediction performance.

ML PROJECT RECAP

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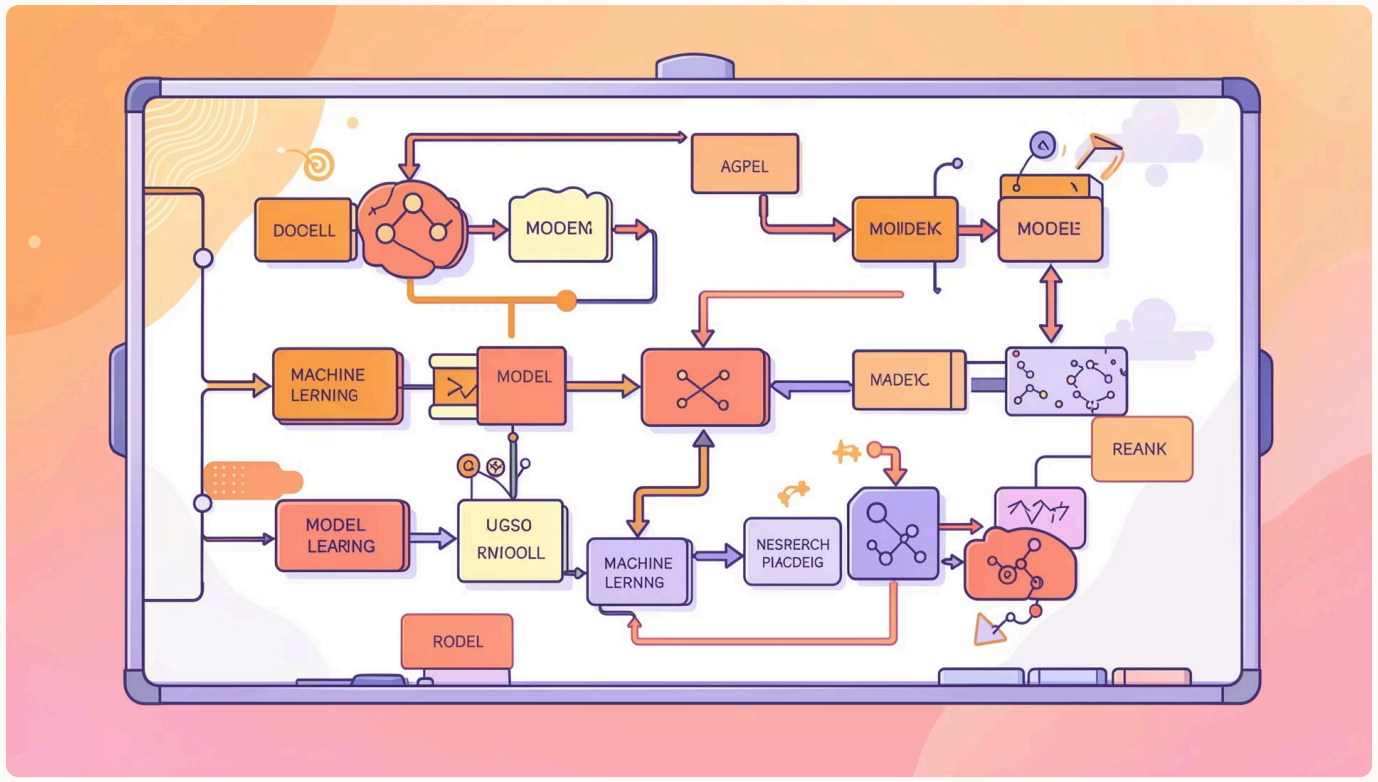
Project Scope & Evolution



Original Scope

Standard comparison of Ridge, Lasso, and nonlinear regression models.

Refined Question

When are linear models insufficient? Does combining linear and nonlinear models improve performance?

Hypothesis

Hybrid or stacking models would outperform individual models.

Standard Benchmarking

Compare Ridge, Lasso, nonlinear models

Residual Learning

Attempted residual nets (failed)

Weighted Hybrid & Stacking

Combine models with weights and stack

The project shifted from simple benchmarking into a deeper study of model expressiveness and ensemble effectiveness.

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Pipeline & Algorithms

1

Preprocessing

Median imputation, one-hot encoding, ordinal mapping for quality features, and log transform on target: **$y = \log(1 + \text{SalePrice})$**

2

Linear Baselines

Ridge: L2 regularization. **Lasso**: L1 regularization. Both trained on standardized features.

3

Gradient Boosting

Sequential residual learning: **$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$** — captures nonlinear relationships and interactions.

4

Hybrid & Stacking

Weighted hybrid: **$\hat{y} = \alpha \cdot \hat{y}_{ridge} + (1-\alpha) \cdot \hat{y}_{GB}$** ($\alpha$=0.05). Stacking uses Ridge + GB predictions as meta-model inputs.

All models evaluated using 5-fold cross-validation and RMSE.

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Preliminary Results

~29.5k

Gradient Boosting

Best-performing model (RMSE)

~51k

Ridge Regression

Underfits nonlinear structure

~57k

Lasso Regression

Excessive sparsity removes useful features

~29.9k

Stacking

No improvement over GB alone

Hybrid and stacking models do not improve over GB — which already captures most predictive structure. Errors increase for higher-priced houses, indicating **heteroscedasticity**.

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Challenges & Solutions

Scope Too Narrow

Introduced hybrid and stacking approaches to deepen the research question.

Complex Preprocessing

Structured imputation, encoding, ordinal mapping, and log transformation resolved skewness and missing data.

Residual Learning Failed

Noisy residuals caused instability — redesigned into a stable weighted hybrid.

Data Leakage Risk

Avoided dynamic tuning of α on validation data to prevent leakage.

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Final Report Plan

Finalize Comparisons

Refine hybrid/stacking explanations and strengthen visual analysis.

Advanced Models

Explore LightGBM/XGBoost, diverse base models, and nonlinear meta-models.

Feature Engineering

Total area, house age, and error analysis across price ranges.

> Our hypothesis — that combining models improves performance — is **not supported**. A well-tuned nonlinear model is often sufficient for structured tabular data.

Key takeaway: **Gradient Boosting alone dominates**. Ensemble methods are theoretically appealing but not always necessary.

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